

Lecture #16

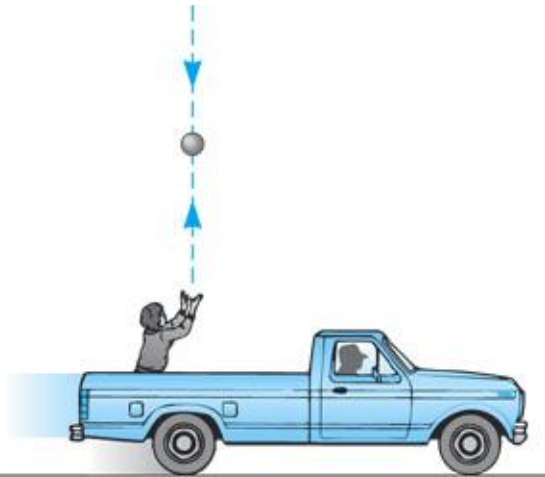
Content to be covered

- ❖ Frames of Reference
 - Inertial Frames
- ❖ Michelson-Morley Experiment
- ❖ Basic Postulates
- ❖ The Galilean Transformation

Resources to be consulted

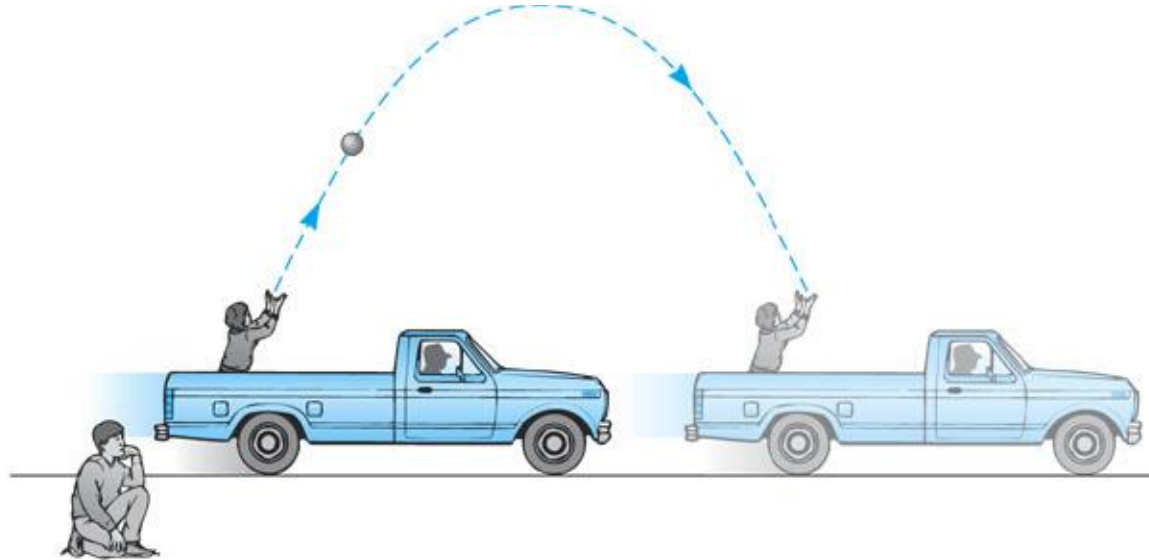
- Concepts of Modern Physics by Arthur Beiser

Experiment at rest



(a)

Experiment in moving frame



(b)

Ball rises and ends up in the thrower's hand. Ball in the air the same length of time.

Experiment looks different from ground observer (parabolic trajectory, speed as a function of time) and observer on the truck.

However, they both agree on the validity of Newton's laws.

Frames of Reference

- ❑ In order to describe the motion of moving bodies, we need to state *where* the object is at a given *time*.
- ❑ But to state *where* an object is, we need to measure its *position* relative to something else.
- ❖ We need a *reference point* from where to define the *position* of objects. Once we have chosen such a point, which is called the *origin*, we can specify the position (x,y,z) , and distance *from the origin*.
- ❖ We also need a *clock* so that we can specify at what *time* '*t*' the object was at the given position.

Inertial Frames

- **Newton's** first law, **the law of inertia**, states that if an object is at rest it will stay at rest if no force is acting on it, and if an object is moving it will keep on moving at constant velocity if no force is acting on it.
- A **frame**, in which this law is correct, is called an **inertial frame**.
- in which no accelerations are observed in the absence of external forces.
- If one such **inertial frame** exists, then an infinite number of other **inertial frames** exist since any **frame** that is moving at a constant relative velocity to the **first inertial frame** is also an **inertial frame**.
- The frames in which the law of inertia **does NOT** hold are those that are accelerating with respect to **inertial frames**. They are called **non-inertial frames**.

Michelson-Morley Experiment

- Experiment designed to measure small changes in the speed of light was performed by Albert A. Michelson and Edward W. Morley.
- Used an optical instrument called an *interferometer* that Michelson invented.
- Device was to detect the presence of the ether.
- Outcome of the experiment was negative, thus contradicting the ether hypothesis.
- A.A. Michelson and E.W. Morley, *American Journal of Science*, 134 – 333, 1887)



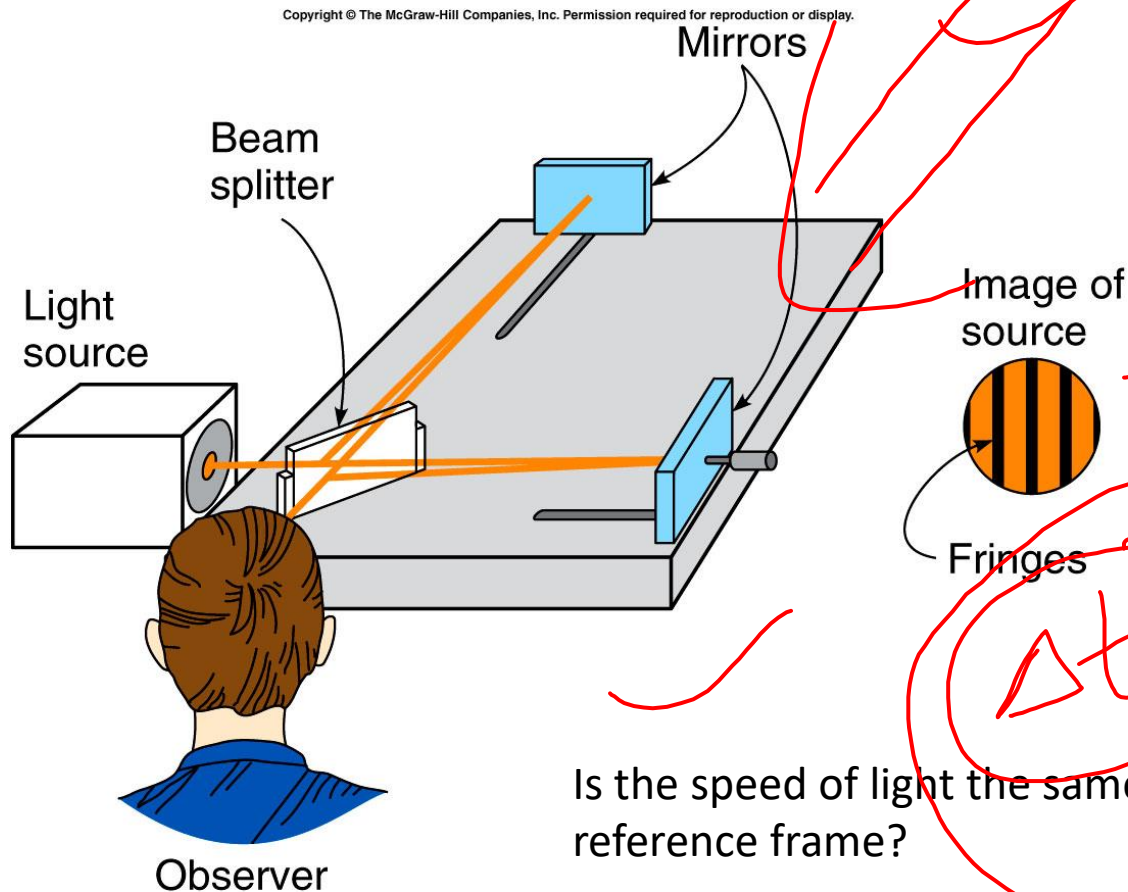
$m \cdot m$
 $=$

~~Electromagnetism~~

~~AT~~
~~HL~~

~~Electromagnetism~~

Michelson-Morley Experiment



Michelson-Morley Experiment

Motion of the Vertical wave:

$$t_1 = 2t = \frac{2L}{\sqrt{c^2 - v^2}} = \frac{2L/c}{\sqrt{1 - v^2/c^2}}$$

For Horizontal wave.

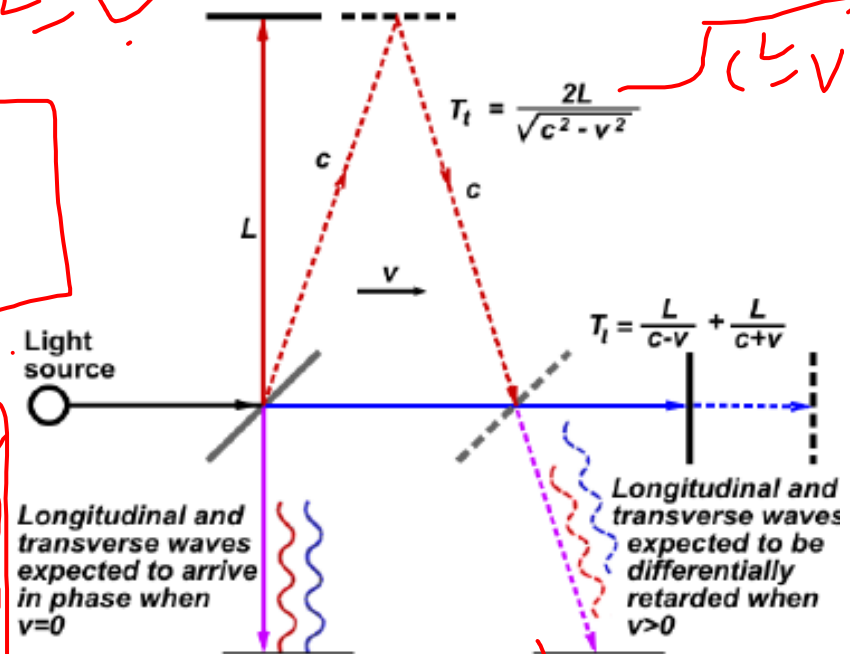
$$t_2 = \frac{L}{c-v} + \frac{L}{c+v} = \frac{2L/c}{(1 - v^2/c^2)}$$

Applying, $y = (1-x)^{-n} \approx 1+nx$, when $x \ll 1$

Thus, difference in time for two waves:

$$t_2 - t_1 = \Delta t = \frac{2L/c}{(1 - v^2/c^2)} - \frac{2L/c}{(1 - v^2/c^2)^{1/2}} = \frac{lv^2}{c^3}$$

Path diff. = $c\Delta t = n\lambda \rightarrow n = c\Delta t/\lambda = lv^2/c^2\lambda$, for n^{th} fringe



$$t_{1a} (P_1 \rightarrow M_1) \quad \text{---} \quad M_2$$

$$t_{1a} = \frac{L_{\text{length}}}{\text{velocity}}$$

$$t_{1a} = \frac{L}{c - v}$$

$$t_{1b} = \frac{\text{length}}{\text{velocity}}$$

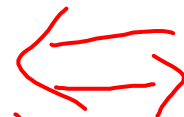
$$t_{1b} \text{ m} \rightarrow P_1$$

$$t_2 = t_{1b} + t_{1a}$$

M_2

M_2

L



2D

$$v_n (v_n)$$

Interferometer

$$v_n' = v_n - v$$

$$v_n' = \frac{c - v}{v_n}$$

$\frac{c}{v_n}$

5 form
alter

$0 \rightarrow v$





Michelson-Morley Experiment

Michelson and Morley performed the experiment in two steps (i) as shown and secondly by turning the apparatus by 90° . Please note that, on turning the apparatus by 90° , the mirror position are changed and now the path diff. becomes:

$$\text{Path diff.} = \frac{lv^2}{c^2}$$

So, on turning the mirror by 90° , the fringe shift = $2lv^2/c^2\lambda$,

For $l = 11\text{m}$, $v = 30000\text{m/s}$, $\lambda = 500\text{nm}$ & $c = 3 \times 10^8\text{m/s}$,

$\Delta n = 0.4$ (This was theoretical result).

Experimentally, there was a shift of 0.01 of a fringe width. This is within the limit of experimental error.

The experiment was repeated at several places at several time in a year, but no shift was observed.

Michelson-Morley Experiment

The negative result had two consequences.

1. The ether doesn't exist and there is no such thing as "absolute motion" relative to the ether.

All motion is relative to a specified frame of reference, not to a universal one.

2. The speed of light is the same for all observers, which is not true of waves that need a material medium in which to occur (e.g. sound & water waves).



The Ultimate Speed

- The speed of light has been defined to be exactly:
 $c = 299\,792\,458\text{ m/s}$
- Light travels at this ultimate speed, as do any massless particles.
- No entity that carries energy or information can exceed this speed limit.
- No particle that does have a mass, can actually reach c .
- Electrons have been accelerated to at least 0.999 999 999 95 times the speed of light—still less than c .

Theory of Relativity

The theory of relativity was developed by Einstein in 1905. It

- deals with the absolute motion of the body through space.
- deals with objects or observers moving relative to each other.

This theory is divided into two parts.

1. Special theory: This theory deals with objects or systems which are either moving at a constant speed or at rest with respect to each other.

2. General theory: Describes the relationship between gravity and the geometrical structure. (not in syllabus)

2 Simple Postulates

- The laws of physics are the same in every *inertial frame* of reference.

– *The Principle of Relativity*

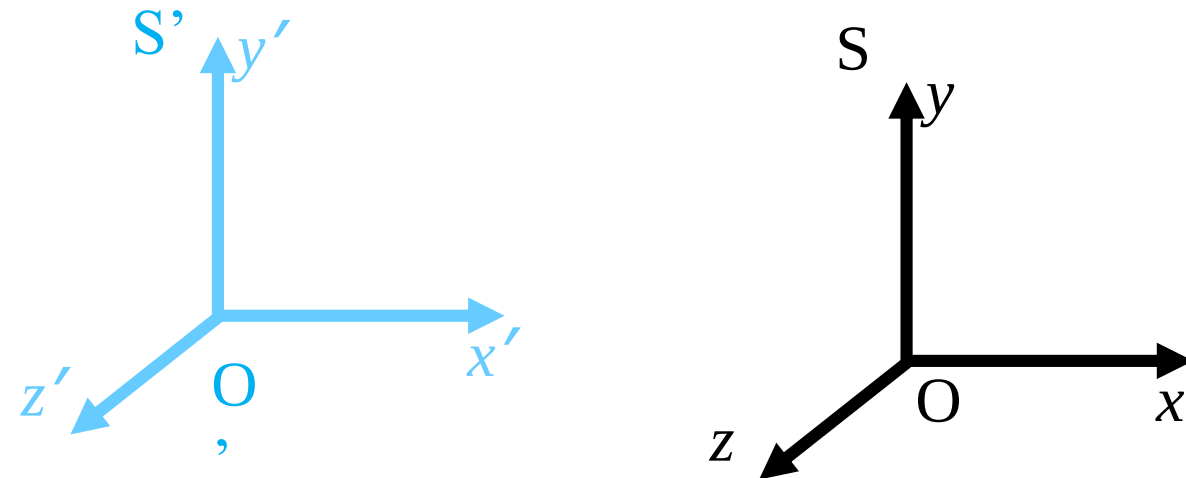
- The speed of light in vacuum/free space is the same in all *inertial frames* of reference, and is independent of the motion of the source.

– *Invariance of the speed of light*

The speed of light $c = 3 \times 10^8 \text{ m/s}$

Newtonian Principle of Relativity

- If Newton's laws are valid in one reference frame, then they are also valid in another reference frame moving at a uniform velocity relative to the first system.
- This is referred to as the **Newtonian principle of relativity** or **Galilean invariance**.



The Galilean Transformation

The equation relating the coordinates of an object in two inertial frames (whose **relative velocity is negligible in comparison to the speed of light**) are called **Galilean transformation**.

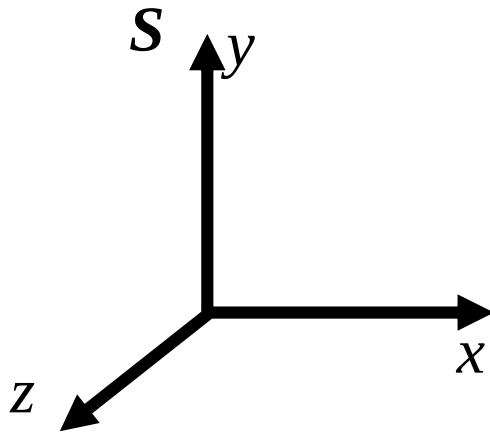
The Galilean Transformation

- For a point P:

- In one frame S: $P = (x, y, z, t)$

- In another frame S': $P = (x', y', z', t')$

- S is at rest and S' is moving with velocity 'v'
- Axes are parallel (for simplicity)
- S and S' are *INERTIAL FRAMES*



$$P = (x, y, z, t)$$

$$P = (x', y', z', t')$$

The Galilean Transformation

S' has a constant relative velocity in the x -direction with respect to S

For a point P

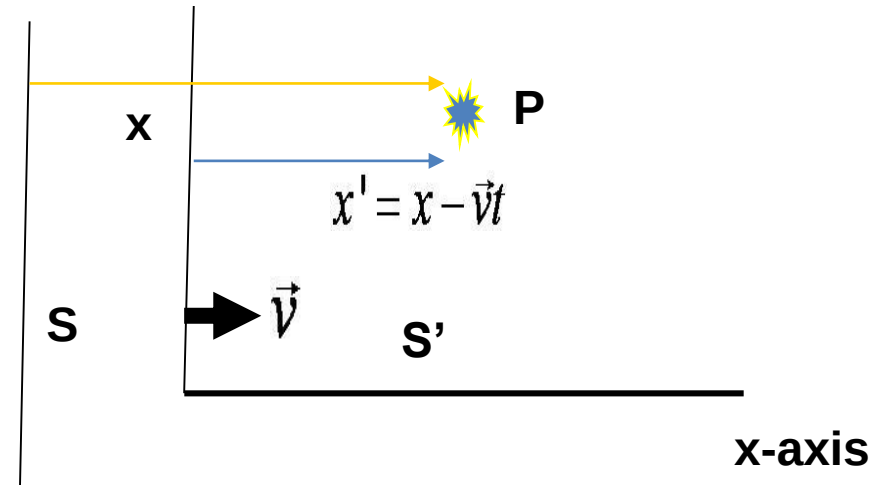
- In system S : $P(x, y, z, t)$
- In system S' : $P(x', y', z', t')$

$$x' = x - \vec{v}t$$

$$y' = y$$

$$z' = z$$

$$t' = t$$



Time (t) for all observers is a *Fundamental invariant*, i.e., the same for all inertial observers.

The Inverse Relations

Step 1. Replace $\vec{v}$ with $-\vec{v}$

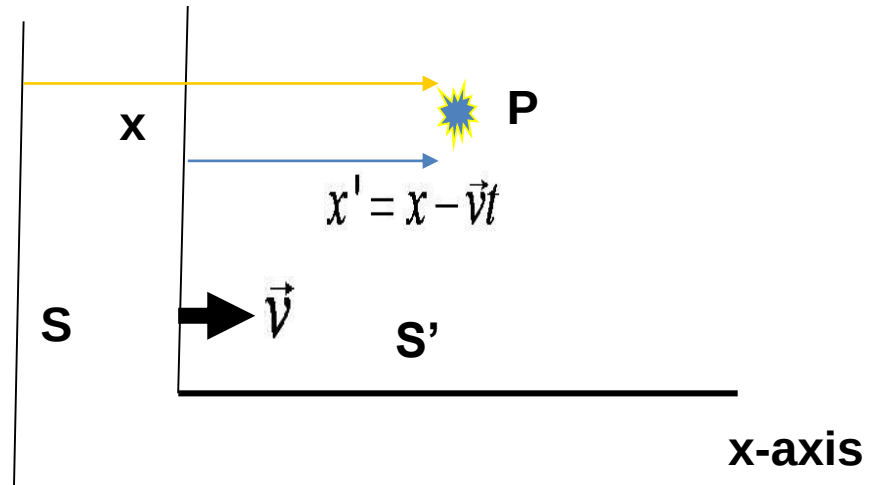
Step 2. Replace “primed” quantities with
“unprimed” and “unprimed” with “primed.”

$$x = x' + \vec{v}t$$

$$y = y'$$

$$z = z'$$

$$t = t'$$



Problem-1

An observer in the laboratory sees two particles colliding at $x = 20.5 \text{ m}$, $y=0$, $z=0$ and $t= 7.2 \text{ s}$. What are the co ordinates of this event in a frame moving at 30.5 m/s with respect to the laboratory frame?

Ans: $x'=-199.1 \text{ m}$, $y'=0$, $z'=0$ and $t'=7.2\text{s}$.

